Jai Hebel (M.Eng)

AI Strategist & Data Scientist

Data scientist and AI specialist with a background in advisory, education, and building custom AI applications. Experienced in AI strategy, agents, data analytics, software, web & app development, machine learning, biomedical research, medical imaging.

WORK EXPERIENCE

Chief AI Scientist (Technical Co-Founder) reena.ai | March 2025 - Now

- Agentic Voice Platform for Dental Operatories (OpenAI Agent SDK, Twilio, 11Labs) with practice management integration, scheduling automation, follow-ups, patient education, and an integrated hardware platform.
- Head of AI strategy and system architecture for an early-stage healthtech startup, overseeing decisions on tooling, data governance, model development, privacy, scalable deployment, and AI innovation.

AI Instructor (Cybersecurity) Info-Tech Research Group | Feb '24 - Now

- AI Strategy Training and Advisory, providing 1:1 coaching to high-profile C-Suite executives on translating complex AI concepts (including custom builds) into actionable AI strategy deliverables for enterprise adoption.
- Co-developed the Agentic Rapid Prototyping Service Platform (OpenAI Agent SDK), a robust agentic platform for facilitators to help clients build custom agentic solutions for niche use cases in a rapid 4-day workshop.
- Designed and delivered the AI Workforce Development Program, training hundreds of IT professionals (including >10,000 employees for the US Department of Labor) on AI foundations, governance, data, strategy, risk management, and responsible AI as the sole instructor.

Senior Software Engineer (Contract) BPK | Aug '23 - Dec '23

- Spearheaded a cutting-edge LLM-powered Conversational Voice Bot (Twilio, HuggingFace Models), innovating patient benefit verifications by autonomously liaising with dental insurance companies over phone, cutting time and costs for reception protocols.
- Prototyped the interactive conversation agent, employing real-time audio transcription over websockets with OpenAI API integration and advanced text-to-speech, producing natural LLM-powered conversation.
- Engineered a hosted Flask server integrated with Twilio API for orchestrating conference sessions, live call streaming, and experiment monitoring, controlled by a React web console and command-line tool.

Data Scientist (Contract) Velmeni AI | Mar '23 - Aug '23

- Trained Visual Transformer networks (UNETRs) for 3D cyst and TMJ segmentation in dental CT images as a new feature in a clinical diagnostic and treatment planning platform.
- Coded data ingestion, preprocessing, and sliding-window patch orchestration of 3D Cone Beam CT data, for hosted dataset management and experiment tracking to produce custom optimized models.
- Deployed containerized production models for GUI app, achieving 88% mean DICE accuracy.

CONTACT

- San Francisco, CA (Open to Remote)
- +1 (480) 717-9916
- jai.hebel@gmail.com
- linkedin.com/in/jkhebel
- github.com/jkhebel

KEY SKILLS

Software Development:

- OpenAI, Langchain, LlamaIndex
- Keras, TensorFlow, PyTorch
- Numpy, Pandas, Scikit-Learn
- Python, C++, Java, MATLAB, R
- JavaScript, React, Node.js, PHP
- MySQL, PostgreSQL, Neo4j
- AWS, GCP, Azure, Linux, Spark

AI and Data Science:

- AI Strategy & Responsible AI
- Agentic Prototyping (OpenAI)
- LLMs (Explainable Systems)
- Conversation Agents (Telephony)
- Computer Vision Deep Learning
- AI Research and Development
- Biomedical AI Applications
- Narrow AI Algorithms
- Statistics, Analytics, Visualization

EDUCATION

Insight Data Science
AI Fellow, Machine Learning
Postgraduate industry fellowship
San Francisco, CA — May 2020

McGill University
M.Eng Biomedical Engineering
Neuroimaging & Surgical Technologies
Thesis: *Current state-of-the-art for glioblastoma segmentation using deep convolutional networks*
Montreal, QC — Dec 2019

Arizona State University
BSE Biomedical Engineering & BSc Psychology
Visuomotor Learning Laboratory
Tempe, AZ — May 2016

Data Engineer Chan Zuckerberg Biohub | Aug '22 — Mar '23

- Designed a deep learning pipeline aiming to track cells in 3D+time microscopy images, as a critical foundation to monitor individual cell maturation for novel developmental biology research.
- Piloted a data conversion pipeline for spatial transcriptomics samples, enabling our team to freely investigate new research on previously inaccessible data formats.
- Supported new product development of a centralized organization-wide hub for experiment tracking and data aggregation by preparing data portals, submission forms, and multi-modal data management.
- Upgraded existing python toolkit from 2D segmentation of cellular images with a heat diffusion model in 3D and 4D images, facilitating computer vision research in microscopy.

Machine Learning Engineer TruTag Technologies | Aug '20 — Feb '22

- Designed custom computer vision machine learning models and software pipelines to be sold with hyperspectral imaging hardware to customers as a comprehensive imaging solution.
- Drove sales by piloting studies (collecting, analyzing, validating, and visualizing customer sample data) to prove that our custom imaging solution achieved success metrics for our customers.
- Applied statistical and machine learning analysis to improve the success rate of our studies, enabling major contracts with several multi-billion-dollar organizations (Kraft, Driscoll's, PepsiCo).
- Expanded our reach to new product lines by designing and implementing deep learning solutions (AWS) to read and decode our novel hyperspectral tags with just a smartphone.
- Built a Python toolkit for data handling and translating analysis methods to our custom hyperspectral image data (~10GB/image) to accelerate project timelines.
- Extended product features to meet customer requests by collaborating with our C++ application engineer to translate popular regression and clustering algorithms into our GUI application.

Software Developer (App & Web) My Dental Hub | Jan '11 - Aug '13

- Improved and developed new databases and API functions for an existing web backend to optimize efficiency and UX (PostgreSQL).
- Designed and developed separate iOS, Android, and Playbook applications, enabling customers to take My Dental Hub software to their mobile smartphones to manage their data on demand.
- Coded and launched a weblog subdomain for company news using PHP and PostgreSQL.

Teaching Assistant (Full-Time) University of Waterloo | Sept '11 - Dec '13

- Organized, wrote, and ran classes for over 100 students under supervision of the department head, provided personal tutoring sessions, and mentored students on their first-year capstone robotics projects.
- Instructed first-year students on introductory and object-oriented programming in C++, and ran lab sessions in Chemistry, Physics, Calculus, Linear Algebra, and Engineering Drafting.

ACADEMIC RESEARCH

AI Fellow (Machine Learning)
Insight Data Science

- Developed a deep learning tool to expedite drug discovery before pre-clinical trials for a 3rd party startup by deploying a variational autoencoder to extract cellular features and predict drug class.
- Classified compounds as cytotoxic using convolutional networks from 4D multi-channel input images of dyed cells and the target compound.
- Extracted cell network features by compressing images to a latent vector space and clustering feature vectors using t-SNE for unsupervised drug classification.

M.Eng Biomedical Engineering
Neuroimaging & Surgical Technologies, McGill University

- Increased accuracy in brain tumor resection research using deep learning by segmenting glioblastoma from multimodal magnetic resonance images.
- Designed and implemented convolutional encoder-decoders to compare with top performers.
- Validated state-of-the-art (DeepMedic: 71.5% DICE; nnUNet: 91% DICE) on common data to show improved performance when emphasis is placed on training algorithm rather than model architecture.

BSE Biomedical Engineering
Visuomotor Learning Laboratory, Arizona State University

- Prototyped and designed a sensation-enhancing sleeve that used electrical stimulation (50μA, 1kHz broadband Gaussian white noise) to improve motor performance in stroke ($p < 0.05$).
- Assisted in the implementation of a novel 3D augmented-reality rhesus monkey limb tracked to the subject's movements for assessing the role of the perceptual field in motor control.